MIL-G-0016265C(SHIPS)
31 August 1962
SUPERSEDING
MIL-G-0016265B(SHIPS)
30 April 1958
USED IN LIEU OF
MIL-G-16265A(In Part)
17 April 1952

MILITARY SPECIFICATION

GASKETS, METALLIC-ASBESTOS, SPIRAL WOUND

(FOR NAVY FLANGED JOINTS IN PIPING SYSTEMS)

This limited coordination military specification has been prepared by the Bureau of Ships based upon currently available technical information, but it has not been approved for promulgation as a revision of Military Specification MIL-G-16265A(in part). It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope.- This specification covers gaskets of Navy design for use in Navy flanged joints in steam piping systems for pressures up to and including 600 pounds per square inch (p.s.i.) and temperatures up to and including 875 degrees Fahrenheit (°F.).

1.2 Classification.- Gaskets shall be of the following series, as specified (see 6.1):

Pipeline flange gaskets, with outer metal rings.
- Series 150.
- Series 300.
- Series 400.
- Series 600.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
- PPP-B-566 - Boxes, Folding, Paperboard.
- PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
- PPP-B-585 - Boxes, Wood, Wirebound.
- PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
- PPP-B-601 - Boxes, Wood, Cleated-Plywood.
- PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
- PPP-B-636 - Box, Fiberboard.
- PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
- PPP-B-676 - Boxes, Set-Up, Paperboard.
- PPP-C-843 - Cushioning Material, Cellulosic.

MILITARY
- MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
- MIL-P-17667 - Paper; Wrapping, Chemically Neutral.
- MIL-F-17823 - Flanges, Pipe, Carbon Steel, 300 PSI, W.S.P. (for Naval Shipboard Use).
- MIL-F-20670 - Flanges, Pipe, Carbon Steel, 150 PSI, W.S.P.
- MIL-G-21032 - Gaskets, Metallic-Asbestos, Spiral Wound (For ASA Commercial Flanged Joints in Piping Systems).

FSC 5330

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

BUREAU OF SHIPS

B-106 - Flanged Cast Steel Pipe Fittings.
B-152 - Flanged Cast Steel Pipe Fittings.
B-174 - Flanges, Steel, Spherical Back Van Stone.
B-175 - Flanged Cast Steel Pipe Fittings.
B-196 - Steel Flanges for Steel Tubing, Welding Neck.
5000-S4823-64496 - Fittings, Forged Steel Socket Welding and Flanged Nipples.
5000-S4824-1385571 - Valves, Cast Steel, Flanged, Globe, Angle and Cross.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N.Y.)

AMERICAN SOCIETY FOR TESTING MATERIALS

A 167 - Specification for Corrosion-Resisting Chromium Nickel Steel Plate, Sheet and Strip.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Qualification.- Gaskets furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Design.-

3.2.1 Pipe line flange gaskets with outer metal rings.- Gaskets shall consist of a refill snapped in a solid metal outer ring with a sliding fit. The ring shall have a thickness of 1/8 ± 0.006 inch. The ring shall have a V-shaped groove or other satisfactory retaining feature machined on the inside periphery to fit the outside periphery of the refill. The design shall be such that the gasket can be compressed without damage to 0.119 inch thickness. Gaskets shall conform to tables I and II.

3.2.1.1 Refills for all series gaskets.- Refills shall be of the inside and outside dimensions specified in tables I and II.

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Table I - Gasket characteristics for series 150 Navy flanges-MIL-F-20670. 1/

Iron pipe size (i.p.s.)	Pipe outside diameter (o.d.)	Gasket refills (see figure 1)		Minimum number of plies Metal 5/			Outer 3/ metal ring o.d.	Test load 4/ gasket pressure
Inches	Inches	I.d. inches 2/	O.d. inches 2/	Inside periphery ±1/2 turn	Outside periphery ±1/2 turn	Total (min.)	Inches	Pounds
1/4	0.540	3/4	1- 3/32	3	2	8	1- 5/8	11,340
3/8	.675	7/8	1- 7/32	3	2	8	1- 3/4	11,340
1/2	.840	1	1-11/32	3	2	8	1-15/16	11,340
3/4	1.050	1-1/4	1-21/32	3	2	9	2- 3/16	15,120
1	1.315	1-1/2	1-31/32	3	2	10	2- 5/8	15,120
1-1/4	1.660	1-3/4	2- 7/32	3	2	10	2- 7/8	15,120
1-1/2	1.900	2-1/4	2-23/32	3	2	10	3- 7/16	22,680
2	2.375	2-3/4	3- 9/32	3	2	10	3-15/16	22,680
2-1/2	2.875	3-3/8	3-31/32	3	2	12	4- 1/2	22,680
3	3.500	3-7/8	4-15/32	3	2	12	5	30,240
3-1/2	4.000	4-3/8	4-31/32	3	2	12	5- 9/16	30,240
4	4.500	4-7/8	5-15/32	3	2	12	6- 1/16	30,240
4-1/2	5.000	5-3/8	6- 1/32	6	3	16	6- 9/16	37,800
5	5.563	5-7/8	6-19/32	6	3	16	7- 3/16	60,600
5-1/2	6.120	6-1/2	7- 7/32	6	3	16	7-11/16	60,600
6	6.625	7	7-25/32	6	3	17	8- 1/4	72,720
7	7.625	8	8-25/32	6	3	17	9- 3/8	72,720
8	8.625	9	9-27/32	6	3	18	10- 7/16	84,840
9	9.625	10	11- 3/32	6	3	22	11- 5/8	126,896
10	10.750	11	12- 3/32	6	3	22	12-11/16	135,960
11	11.750	11-1/2	12-19/32	6	3	22	14- 1/4	145,024
12	12.750	12-1/2	13-25/32	6	3	25	15- 5/16	163,152
14 o.d.	14.000						16- 1/2	238,830
15 o.d.	15.000						17- 1/2	251,400
16 o.d.	16.000						18- 9/16	251,400
18 o.d.	18.000						20- 5/8	276,540
20 o.d.	20.000						22-11/16	396,120

1/ Gasket for 150 p.s.i., 1050°F. steam service when used with Specification MIL-F-20670.
2/ Tolerance ± 1/64 inch on the inside diameter and plus 1/32 inch or minus 1/32 inch on the outside diameter.
3/ I.d. of outer metal ring equals o.d. of refills plus 1/32 inch for all sizes.
4/ Corresponds to bolting load of 30,000 p.s.i. unit stress. Bolts; 3/4 inch diameter, 10 threads; 7/8 inch diameter, 9 threads; all other bolts 8 threads.
5/ There shall be no asbestos plies between the metal plies on the inside and on the outside periphery.

3.3 Materials.-

3.3.1 Metal strip.- The metal strip used in the gasket refills shall be ONE of the following materials:

(a) Corrosion resisting steel, AISI type 347, cold rolled and annealed, in accordance with ASTM A167.
(b) Corrosion resisting steel, AISI type 316L, in accordance with ASTM A176.
(c) Corrosion resisting steel, AISI type 309S, in accordance with the applicable requirements of ASTM A176, except that type 309S shall be stabilized with columbium (and tantalum not exceeding 0.10 percent) and their combination to be not less than 10 times the carbon content.

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Table II - Series 300, 400, and 600 Navy flanges.

I.p.s.	Pipe o.d.	Gasket refills (see figure-1)		Minimum number of plies			Outer metal ring[2] o.d. inch			Test load[6] gasket pressure
				Metal[1]			[3] Series 300	[4,8] Series 400	[5,8] Series 600	
Inches	Inches	I.d. inches[1]	O.d. inches[1]	Inside periphery ±1/2 turn	Outside periphery ±1/2 turn	Total (min.)				Pounds
1/4	0.540	1/2	27/32	4	3	9	2- 1/8	1- 3/4	1- 3/4	12,600
3/8	.675	3/4	1- 7/32	6	3	12	2- 3/8	2- 1/8	2- 1/8	12,600
1/2	.840	3/4	1- 7/32	6	3	12	2- 3/8	2- 1/8	2- 1/8	12,600
3/4	1.050	1	1-17/32	6	3	14	2-11/16	2- 5/8	2- 5/8	20,200
1	1.315	1- 1/4	1-27/32	6	3	14	3- 1/8	2- 7/8	2- 7/8	20,200
1-1/4	1.660	1-11/16	2-11/32	6	3	14	3- 7/16	3- 1/4	3- 1/4	24,200
1-1/2	1.900	2	2-23/32	6	3	15	4	3- 3/4	3- 3/4	36,200
2	2.375	2- 9/16	3-11/32	6	3	15	4- 1/2	4- 3/8	4- 5/16	48,400
2-1/2	2.875	2-15/16	3-27/32	6	3	16	5- 3/16	5- 1/8	5- 1/16	72,500
3	3.500	3- 3/4	4-23/32	6	3	18	5- 3/4	5- 7/8	5-13/16	72,500
3-1/2	4.000	4- 1/8	5- 7/32	6	3	20	6- 5/16	6- 5/16	6- 5/16	100,600
4	4.500	4- 3/4	5-27/32	6	3	20	6- 7/8	6-15/16	6-15/16	100,600
4-1/2	5.000	5- 5/16	6-15/32	6	3	22	7- 7/16	7- 9/16	7- 7/16	132,200
5	5.563	5-13/16	6-31/32	6	3	22	8	8- 5/16	8- 3/16	132,200
5-1/2	6.125	6- 3/8	7-19/32	6	3	22	8-11/16	8-13/16	8-15/16	150,900
6	6.625	6- 7/8	8- 7/32	6	3	24	9- 1/4	9-11/16	9- 9/16	198,400
6-1/2	7.125	7- 7/16	8-23/32	6	3	24	9- 7/8	10- 5/16	10- 3/16	198,400
7	7.625	7-11/16	9- 7/32	6	3	24	10- 7/16	10-13/16	10-11/16	262,000
7-1/2	8.125	8- 5/16	9-27/32	6	3	27	11- 1/8	11- 5/16	11- 5/16	262,000
8	8.625	8- 7/8	10-11/32	6	3	27	11-11/16	11-15/16	11-13/16	262,000
8-1/2	9.125	9- 5/16	10-27/32	6	3	27	12- 1/4	12- 5/16	--------	262,000
9	9.625	9- 7/8	11-15/32	6	3	32	12-13/16	12-13/16	13- 7/16	334,400
9-1/2	10.250	10- 5/16	11-31/32	6	3	32	13- 3/8	13- 9/16	--------	349,300
10	10.750	10-13/16	12-15/32	6	3	32	13-15/16	14- 1/16	15-11/16	445,900
10-1/2	11.250	11- 5/16	12-31/32	6	3	32	14- 1/2	14- 9/16	--------	349,300
11	11.750	11- 3/4	13-23/32	6	3	32	15- 3/16	15- 7/16	16-11/16	557,400
11-1/2	12.250	12- 5/16	14- 7/32	6	3	32	15- 3/4	15-15/16	--------	445,900
12	12.750	12- 7/8	14-23/32	6	3	32	16- 3/8	16- 7/16	17-15/16	557,400
14 o.d.	14.000	14- 1/4	15-31/32	6	3	32	17- 1/2	18-15/16	--------	557,400
15 o.d.	15.000	15- 1/4	16-31/32	6	3	32	18- 3/4	20- 3/16	--------	557,400
16 o.d.	16.000	16- 1/4	18- 7/32	6	3	35	19-13/16	21- 1/16	--------	577,000

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1/ Tolerance plus or minus 1/64 inch on the inside diameter and plus 1/32 or minus 1/32 on the outside diameter.
2/ I.d. of outer metal ring equals o.d. of refills plus 1/32 inch for all sizes.
3/ Gaskets for 300 p.s.i. steam service when used with Drawing B-106 or 5000-S4824-1385571 and Specification MIL-F-17823.
4/ Gaskets for steam service pressure 301 to 400 p.s.i. gage and 850°F. Maximum temperature and boiler feed service pressure 401 to 550 p.s.i. and 425°F. maximum temperature, when used with Drawing B-152 or 5000-S4823-64496.
5/ Gasekts for steam service pressure 401 to 600 p.s.i. gage and 850°F., maximum temperature and boiler feed service pressure 551 to 750 p.s.i. gage and 425°F., maximum temperature, when used with Drawing B-196, B-174, B-175, 5000-S4823-64496 and valves for 600-pound service.
6/ Corresponds to bolting load of 25,000 p.s.i. unit stress for 1/4 to 1 inch inclusive and 30,000 p.s.i. for 1-1/4 inch and above.
7/ ~~There shall be no asbestos plies between the metal plies on the inside and on the outside periphery.~~
8/ Series 400 and 600 Navy flange gaskets for sizes 1/4 to 3 inches inclusive are interchangeable with ASA flange gaskets (Specification MIL-F-2103) for series 300, 400, and 600 for the same size.

MIL-G-21032B

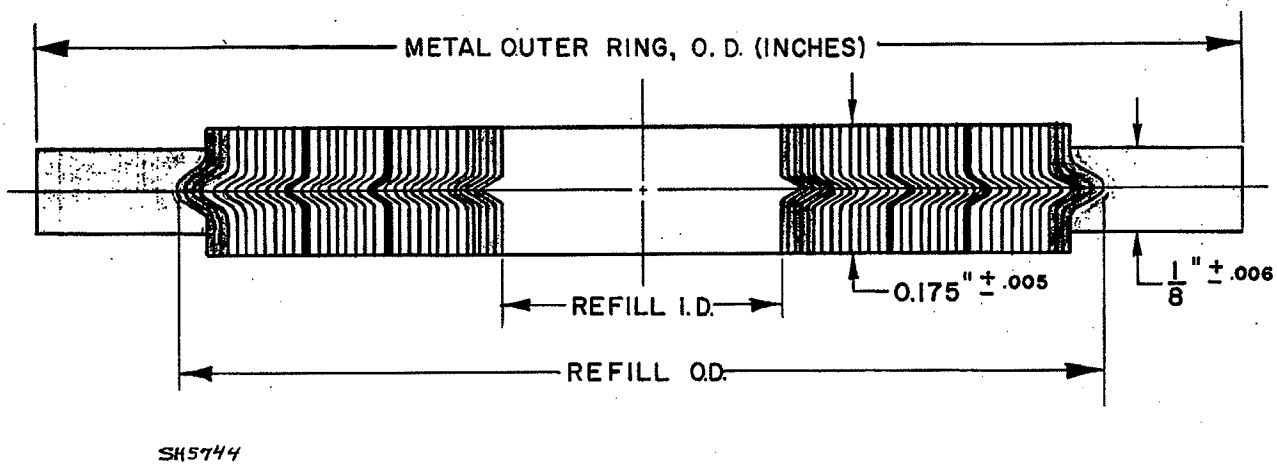


Figure 1 - Gasket.

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3.3.1.1 Special application. - For special applications other materials shall be specified or specifically approved by the bureau or activity concerned.

3.3.2 Metal outer ring. - The metal outer ring shall be of soft carbon steel. At the option of the manufacturer a protective coating may be applied to the metal outer ring. The resulting dimensions shall conform to tables I and II.

3.4 Construction. - The gaskets shall be composed of a single strip of metal wound in spiral dovetail shape, starting on the inside and working outward ply by ply to the desired size. Between some of the plies shall be a cushion of asbestos paper (see tables I and II) either a continuous single length or in two lengths depending upon the design. The asbestos paper may be either a single strip or in two strips. Spot welds on the gasket inside diameter shall be not less than three in number and approximately spaced at 2 inches maximum spacing. The outside weld shall consist of not less than three spot welds.

3.4.1 Service. - Complete gaskets and refills specified in tables I and II shall be satisfactory for temperatures up to 1050° F.

3.4.2 Metal strip. - The metal strip shall be not less than 0.007 and not more than 0.009 inch in thickness.

3.4.3 Gasket thickness. - The thickness of the finished gasket (before compression) shall be 0.175 ±0.005 inch.

3.4.4 Plies. - A ply shall consist of one 360 degree turn of metal strip. Plies shall be counted adjacent to a terminal weld. An asbestos ply shall consist of one 360 degree turn of asbestos strip(s) depending upon the design.

3.4.5 Gaskets shall be capable of passing the compression load tests indicated in 4.5.1.

3.5 Marking. -

3.5.1 Each gasket shall be steel stamped clearly on one side of the outer ring to show the manufacturer's name, pipe size, the pressure series and "NAVY MIL" to indicate that the gasket is for Navy flanges and also meets the requirements of this specification (example - "2-600 NAVY-MIL"). For interchangeable gaskets (sizes 1/4 to 3 inches, inclusive) as specified in footnote 8 of table II, the marking for pressure series and "NAVY-MIL" shall be as follows: "4/600 N :3/4/600 ASA-MIL".

3.5.2 Refills. - When refills are purchased separate from the rings, each refill shall be identified by attaching a gum label or by other means as approved by the bureau or activity concerned, having the manufacturer's name, pipe size, pressure series and N printed thereon. For sizes above 8-inch nominal pipe size, they may be kraft paper wrapped, enclosed in paper or plastic bag and sealed with gum tape. For all other sizes, they shall be placed in a plastic bag or envelope and sealed with gum tape.

3.6 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests.[1/] - Qualification examination and testing of gaskets shall be conducted at a laboratory satisfactory to the Bureau of Ships. A supplier may have his gaskets qualified in any pressure series by passing the specified tests for the series concerned. Qualification examination and tests shall consist of those specified in 4.4 and 4.5.1.

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

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4.2.1 Samples for qualification tests. - For purposes of qualification testing, three gaskets of each of the following series and sizes shall be furnished:

Pressure series	Sample sizes required
2/ 150	2 inch, 8 inch, 12 inch
300, 400 and 600	3 inch, 6 inch, 10 inch

4.3 Sampling for quality conformance inspection. -

4.3.1 Lot. - All gaskets of the same series and size offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.3.2 Sampling for visual and dimensional examination. - Sample gaskets shall be selected at random from each lot in accordance with table III for the examination specified in 4.4.

Table III - Sampling for visual and dimensional examination.

Lot size (number of gaskets)	Sample size (number to be examined)	Acceptance number (Defectives)	Rejection number (Defectives)
2 to 10	All	-	-
11 to 25	10	0	1
26 to 40	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 to 500	50	2	3
501 to 800	75	3	4
801 to 1300	110	4	5
1301 and over	150	5	6

4.3.3 Sampling for compression test. - Sample gaskets shall be selected from each lot in accordance with table IV for the tests specified in 4.5.1. If any gasket fails the compression test, this shall be cause for rejection of the entire lot.

Table IV - Sampling for compression test.

Lot size (Number of gaskets)	Sample size (Number to be tested)
1 to 25	1
26 to 180	2
181 to 500	3
501 to 800	5
801 to 1,300	7
1,301 to 3,200	10
3,201 to 8,000	15
8,001 to 22,000	25
22,001 and over	35

2/ Since the gasket body (or refill) is identical for series 300, 400 and 600, only one set of samples for these series is required.

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4.4 Visual and dimensional examination. - Each of the sample gaskets selected in accordance with 4.3.2 shall be examined and measured to verify conformance with all of the requirements of this specification which do not include the compression tests. Examination shall be conducted as specified in table V. The number of plies shall be counted, a magnifying glass being used to insure accuracy of the count.

Table V - Classification of defects in accordance with Standard MIL-STD-105.

Categories	Defects
Critical:	None defined.
Major:	
101	Type of gasket not specified.
102	Number of metal plies less than specified minimum.
103	Spot welds (on inside and outside diameters) missing, not the required number, incorrectly spaced, or spot welds defective.
104	Thickness of metal strip not within the specified maximum and minimum values, or metal strip not continuous (one piece).
105	Thickness of finished gasket not within specified tolerances.
106	Gasket refill not a sliding fit in the metal ring, or metal ring not provided with a satisfactory means of retaining the refill.
107	Inside or outside diameter of refill not within the specified tolerances.
108	Identification markings missing, illegible, incorrect or not as specified.

4.5 Test procedures. -

4.5.1 Compression load test. -

4.5.1.1 The test shall consist of subjecting the gaskets to compression in a Baldwin Southwark, or equal, hydraulic compression machine of suitable capacity and measuring the thickness of the gaskets while under the tests loads specified in tables I and II, as applicable, for gaskets.

4.5.1.2 The gaskets shall be tested between steel test plates, the surfaces of which shall have a smooth finish machined with a circular lay (concentric or spiral) having a minimum of 32 serrations per inch of face width with depth of serrations not exceeding 0.003 inch of an RHR finish of from 250 to 500. For special installations involving radioactive service or hazardous fluids where a finer finish is required, and so indicated in the contract or order, a maximum RHR of 125 shall be used. The test plates shall be centrally located under the ball and socket head of the test machine. The ball and socket head shall be used to equally distribute the load applied to the gasket. By means of inverted T straps, the upper compression plate shall be suspended and held in place when inserting a gasket. Four dial indicators graduated in 1/1000 inch divisions, located 90 degrees apart and mounted on the upper compression plate are employed for measuring the thickness of the gasket, or other mechanical means as approved by the bureau or activity concerned.

4.5.1.3 During the test, the increment of load shall be applied until a load corresponding to the test load in accordance with tables I and II is obtained for the gasket under test. The maximum rate of loading allowed will be 2,000 pounds per second.

4.5.1.4 The steel test plates shall be placed in contact with each other and under an applied load of 3,000 pounds, the dial indicators shall be set so that a zero reading is obtained with the maximum travel of the indicator plunger. The plates shall then be separated, a gasket centrally located between plates and the upper plate brought into contact with the gasket but no load applied. The dial indicators shall then be read, and the average of the four readings should closely check the original micrometer thickness of the gasket. The load shall then be applied and the rate of loading depends on the size of gasket and the maximum load to be applied. The specified load shall be maintained only for a sufficient time, 5 to 10 seconds, to obtain the dial indicator readings. By means of reading the dial indicator scales counter clockwise instead of clockwise the actual thickness of the gasket shall be directly obtained. The average of the four dial indicator readings shall be considered to be the thickness of the gasket. After completion of the test, the load shall

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be released but the upper compression plate shall be left in contact with the gasket for approximately 10 seconds and the dial indicator readings shall be taken to determine the recovery of the gaskets. The test plates shall then be separated, the gasket removed from the test machine, measured for thickness by micrometers, examined for buckling on the inner periphery, condition of welds, and binding of the gasket filler in the outer metal ring. It may be noted that the final micrometer thickness and the released average dial indicator thickness of the gasket will be approximately the same providing the dial indicators are in proper adjustment. Occasionally between test runs the zero setting of the dial indicators shall be checked with the test plates in contact with one another. No buckling on the inner edge of the gasket shall be allowed. Gaskets shall at least recover 0.010 inch over the compressed value obtained in the compression test specified in 4.5.1.3.

4.5.2 Possible test failures.- Possible test failures are defined as follows:

(a) Gasket buckled excessively on inner periphery to the extent that metal plies are separated.
(b) Welding failed.
(c) Gasket thickness recovered less than 0.010 inch after release of compression test load.

4.5.3 Chemical analysis.- Mill certification of materials may be accepted in lieu of performing chemical and physical tests. Such certification should contain the results of the chemical check analysis of the heat from the steel obtained, the heat number and date. When considered necessary, the samples selected in accordance with 4.2.1 and 4.3 shall be subjected to a chemical analysis to determine conformance with 3.3.1.

4.6 Comparison examination and tests.- Comparison examination and tests shall be conducted to verify continuing quality of the approved product and compliance with the specification requirements. These tests will be on samples representing those series and sizes which were tested for qualification and approved and will be conducted on samples supplied by the manufacturer, at no cost to the Government. Comparison examination and tests are not required more often than every two years unless the inspector has reason to consider it necessary. Comparison examination and tests shall be those specified in 4.4 and 4.5.1 plus any others specifically requested by the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 Level A.- Gaskets and refills, quantity as specified (see 6.1), shall be packaged in boxes conforming to Specification PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676, at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.1.1.1 Bundling.- When specified (see 6.1), gaskets shall be bundled in the quantities and tied as follows:

5.1.1.1.1 Gaskets (with outer metal rings).-

Size (O.D.)	Quantity and tying
Up to and including 5 inches.	25 pieces.
Over 5 inches to 8 inches, inclusive.	25 pieces, tied in 2 equidistant places.
Over 8 inches to 11 inches, inclusive.	25 pieces, tied at 4 equidistant places.
Over 11 inches to 16 inches, inclusive.	10 pieces, tied at 4 equidistant places.
Over 16 inches.	5 pieces, tied at 4 equidistant places.

Wrapping paper shall conform to Specification MIL-P-17667.

5.1.1.2 Bulk.- When specified (see 6.1), gaskets shall be bulk packed directly into shipping containers as specified in 5.2.1 and 5.2.2.

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5.1.2 Level C. - Gaskets and refills shall be packaged in a manner sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2 Packing. -

5.2.1 Level A. - Gaskets and refills, packaged as specified (see 6.1), shall be packed in fiberboard boxes conforming to class 2, grade 3 of Specification PPP-B-636. Alternatively, gasket and refills may be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-585	Class 3 use
PPP-B-591	Overseas type
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-576	Class 2

Shipping containers shall have caseliners conforming to Specification MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to Specification MIL-L-10547. Caseliners for class 2 fiberboard boxes conforming to Specification PPP-B-636 may be omitted provided all center and edge seams and manufacturer's joints are waterproofed in accordance with the appendix to Specification PPP-B-636. Boxes shall be closed, strapped or banded in accordance with the applicable box specification or appendix thereto. The gross weight of wood or woodcleated boxes shall not exceed 200 pounds.

5.2.2 Level B. - Gaskets and refills, packaged as specified (see 6.1), shall be packed in fiberboard boxes having a minimum bursting strength of 275 p.s.i., conforming to Specification PPP-B-636, class 1. Alternatively, gaskets and refills may be packed in containers conforming to any one of the following Specifications at the option of the contractor:

Specification	Type or class
PPP-B-585	Class 1 or 2 use
PPP-B-591	Domestic type
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-576	Class 1

Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

5.2.3 Level C. - Gaskets of like series and size, packaged as specified (see 6.1), shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.2.4 Filling of voids. - Voids or cavities in the shipping containers, caused by wrapped bundles or bulk packing, shall be filled with cushioning materials to prevent shifting of contents and damage during handling, shipment and storage. The use of loose excelsior, shredded newspaper and similar materials which have not been refined for non-corrosiveness shall not be used unless these materials are enclosed in a sealed waterproof barrier material. Cushioning materials shall conform to Specification PPP-C-843 or other materials that are noncorrosive to the packed item.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Series and size of gaskets required.

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(c) If refills are required, specify series and size required.
(d) Quantity per unit package (see 5.1.1).
(e) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 16265, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D.C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

~~6.3 Copies of "Provisions Governing Qualification"~~ may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

6.4 Superseding data. - This specification replaces that part of Specification MIL-G-16265A covering classes b (in part) and c. Classes a and b (in part) of Specification MIL-G-16265A have been replaced by types I and II of Specification MIL-G-21032 and gaskets for commercial flanges should be purchased under Specification MIL-G-21032.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)*. PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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